Design and Analysis of a Stochastic Optimization Model for Day-Ahead Energy Planning of a Solar–Battery Campus Microgrid

21AIE321T – Stochastic Decision Making · Assignment IV (CLA-2) · Mini Project
SRM Institute of Science and Technology, Tiruchirappalli · B.Tech AI, Semester V · AY 2026-2027
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Abstract

A campus microgrid with rooftop solar (600 kWp), a battery (800 kWh) and a grid connection must **commit to hourly day-ahead electricity purchases before tomorrow's sunshine and load are known**; any mismatch is settled in the real-time market at twice the price (shortfall) or resold at a low export price (surplus). We formulate a **two-stage stochastic linear programme** with a 24-hour battery model as recourse, solve it by **Sample Average Approximation (SAA)** with 300 scenarios, add a **mean-CVaR** risk-averse variant, and compare it with a deterministic mean-forecast plan over 2,000 out-of-sample simulated days. The stochastic plan costs **₹33,548/day** versus **₹34,043/day**: a **Value of the Stochastic Solution (VSS) of ₹494.9/day** (95% CI ₹450–541, 1.45% ≈ ₹1.81 lakh/year), with 11% lower cost volatility and 38% less expensive real-time purchasing. The gain is concentrated on overcast days (₹1,549/day) and grows rapidly when real-time prices are high (10.2% at a 4× premium, ≈ 0 at no premium). A stress test shows the plan loses to the deterministic one if load turns out 10% lower than in training (Section 6.3).

1. Problem Selection and Real-World Grounding

Institutions in Tamil Nadu increasingly install rooftop solar and batteries, yet still buy a large share of their energy from the grid under time-of-day tariffs and, for larger consumers, scheduled *day-ahead* procurement with imbalance settlement for deviations. Tomorrow's solar output depends on cloud cover and the monsoon, and the load depends on temperature, academic calendar and occupancy; both are uncertain at the time the purchase schedule must be submitted. This is a textbook **two-stage stochastic decision problem with recourse**: the day-ahead schedule is the *here-and-now* decision; battery dispatch, real-time purchases and resale are *wait-and-see* corrections.

SDG 9 (Target 9.5) link. Target 9.5 asks for enhanced industrial and technological capability and innovation. Stochastic energy scheduling is a software-only upgrade that makes existing solar and storage assets more valuable and reduces price-risk exposure; Section 6.4 measures its impact (cost, real-time purchases, grid import and CO_2), meeting SDG Integration Level 3 (Impact & Measurement). **Data note:** no campus smart-meter data were available, so a 730-day hourly history of PV output and load was *realistically simulated* (Section 2.3) and saved in **data/microgrid_history_hourly.csv**; the code accepts real data in the same format.

2. Formal Problem Formulation

2.1 Decision variables and parameters

Hours t = 0,...,23. **First stage (here-and-now):** $x_t \in [0, G]$, kWh bought day-ahead at known time-of-use price π_t. **Random data** $\omega = (PV_t(\omega), L_t(\omega))$: solar output and load (kW ≡ kWh/h). **Second stage (recourse)** for each ω: u_t real-time purchase at $\rho\pi_t$ (ρ = real-time multiplier), z_t resale at price σ, c_t/d_t battery charge/discharge, e_t state of charge, l_t unserved load at penalty V (value of lost load), w_t curtailed PV; δ is the battery wear cost per kWh discharged.

Parameter	Value	Parameter	Value
PV capacity (kWp)	600.0	Real-time price multiplier	2.0
Battery energy (kWh)	800.0	Export / resale price (INR/kWh)	3.0
Battery power (kW)	200.0	Value of lost load (INR/kWh)	60.0
One-way efficiency	0.92	Battery wear (INR/kWh)	1.2
Min SOC	0.1	DA price min/max (INR/kWh)	5.0 / 9.5
Grid connection (kW)	450.0		

Table 1. System and market parameters. Day-ahead tariff: ₹5.0 (0–5 h), 6.5 (6–9 h), 6.0 (10–17 h), 9.5 (18–21 h), 6.0 (22–23 h).

2.2 Two-stage stochastic programme

Minimise $\Sigma_t \pi_t x_t + E_\omega[\, Q(x, \omega) \,]$, with the recourse cost

$Q(x,\omega) = \min \Sigma_t (\rho\pi_t u_t - \sigma z_t + \delta d_t + V l_t)$

s.t. $x_t + u_t + (PV_t - w_t) + d_t + l_t = L_t + c_t + z_t$ (energy balance)

$e_t = e_{t-1} + \eta c_t - d_t/\eta$, $e_{-1} = e_{23} = 0.5E$, $0.1E \le e_t \le E$, $0 \le c_t, d_t \le P_b$ (battery)

$x_t + u_t \le G$, $0 \le w_t \le PV_t$, $0 \le l_t \le L_t$, $z_t, u_t \ge 0$ (grid, curtailment, shedding limits)

Every day-ahead schedule is feasible for every scenario (resale and shedding always absorb mismatch), so the problem has **relatively complete recourse**. Buying too much day-ahead forces cheap resale (σ = ₹3 vs ₹5–9.5 paid), buying too little forces real-time purchases at 2π: a 24-period, battery-coupled analogue of the newsvendor trade-off. **Risk-averse variant** (Rockafellar–Uryasev): min $(1-\lambda)\cdot E[\text{cost}] + \lambda \cdot \text{CVaR}_{90\%}[\text{cost}]$, the mean cost of the worst 10% of days, linearised with η and $z_s \geq \text{cost}_s - \eta$, $z_s \geq 0$.

2.3 Probability model of the uncertain data

Weather state per day: sunny (55%), partly cloudy (30%), overcast (15%), with daily clearness index $\kappa \sim N(0.88, 0.06)$, $N(0.60, 0.12)$, $N(0.28, 0.10)$ respectively (clipped to [0.05, 1]). $PV(t) = 600\ \text{kWp} \cdot \text{clear-sky shape}(t) \cdot \kappa \cdot (1 + \varepsilon_t)$ where ε_t is an AR(1) process ($\varphi = 0.7$) whose standard deviation is 5%/25%/20% by state, capturing intra-day cloud transients. **Load** = base campus profile (172–410 kW) × day factor $(1 + 0.06(\kappa - 0.6)/0.3) \cdot \text{lognormal}(\sigma = 5\%)$ × hourly AR(1) noise (3%): hot sunny days raise cooling load, so PV and load are positively correlated across days. This process generates the 730-day history (daily PV–load correlation 0.67, CV of daily PV energy 0.36). **The optimiser never sees the true process**: scenarios are drawn by a *smoothed bootstrap* of the historical days, while validation uses fresh days from the true process.

Weather state	Days	Share %	PV kWh/day (mean)	PV kWh/day (std)	Load kWh/day	Mean peak load (kW)
Sunny	383	52.5	3,793	274	7,699	437
Partly cloudy	227	31.1	2,573	681	7,239	410
Overcast	120	16.4	1,171	462	6,786	386
ALL DAYS	730	100.0	2,983	1,075	7,406	420

Table 2. Statistics of the 730-day history. Solar covers on average 40% of campus load, but only 17% on overcast days.

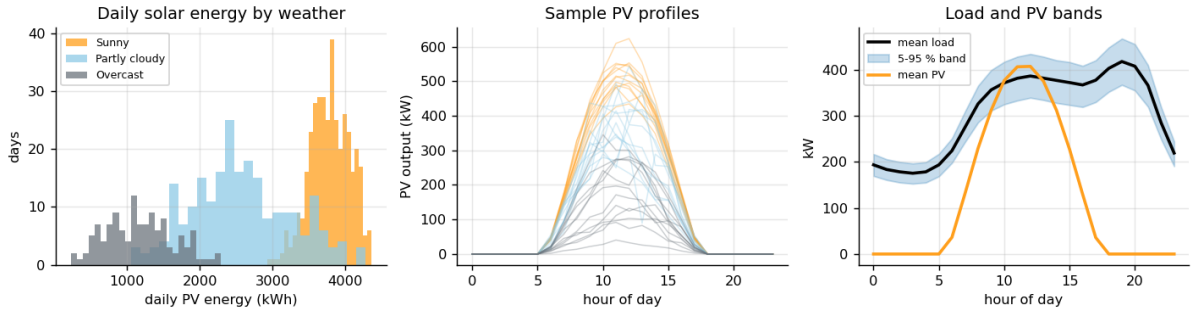


Figure 1. Left: daily PV energy by weather type. Centre: sample PV profiles. Right: mean load and PV with 5–95% load band.

3. Methodology and Justification of the Chosen Approach

Approach: two-stage stochastic programming solved by Sample Average Approximation (SAA). E[Q(x,ω)] is replaced by the average over S sampled days, giving one large LP in extensive form (24 + 168S variables, e.g. 50,424 for S = 300) solved exactly with HiGHS (SciPy). **Why SAA:** (i) the structure is genuinely two-stage with battery-coupled recourse, which stochastic programming represents exactly; (ii) the uncertainty is a 48-dimensional correlated vector (hourly PV and load) with state-dependent noise, impossible to discretise on a small scenario tree but natural to sample; (iii) SAA solutions converge to the true optimum as S grows (Kleywegt et al., 2002; verified empirically in Section 5.4); (iv) the LP gives exact optima, whereas a metaheuristic such as simulated annealing would need an inner LP solve per candidate schedule and gives no optimality guarantee. A chance constraint on supply was not needed because the grid connection always provides a backstop; risk is instead managed through CVaR.

Baseline and metrics. The *deterministic baseline* solves the same model with PV and load fixed at their historical mean profiles (S = 1). EEV = expected cost of that schedule on random days; RP = expected cost of the stochastic schedule; **VSS = EEV − RP**. *Wait-and-see* (WS) is the expected cost if the schedule could be chosen after seeing the day; **EVPI = RP − WS**. **Validation:** every schedule is evaluated by solving the exact recourse LP on 2,000 fresh days from the true process (EVPI on 500 of them), with a 2,000-resample bootstrap confidence interval for VSS.

4. Implementation Summary

Python 3 with NumPy, SciPy (HiGHS), pandas, Matplotlib and ReportLab. **src/microgrid.py** holds the uncertainty model, sparse extensive-form LP builder (risk-neutral and CVaR), exact recourse evaluator, wait-and-see solver and risk metrics. **src/run_experiments.py** runs all experiments end-to-end with fixed seeds and writes 11 result tables (**results/**), 8 figures (**figures/**) and **summary.json**; **src/make_report.py** builds this PDF from them; **tests/test_microgrid.py** checks the LP against an independent exact calculation, energy-balance feasibility, CVaR behaviour and the wait-and-see bound. An executed notebook is in **notebooks/**.

5. Results: Optimal Solution, Expected Value and Risk

5.1 Optimal day-ahead schedule

Hour	DA price ₹/kWh	Mean load kW	Mean PV kW	Det. purchase kWh	Stoch. purchase kWh	Difference
0	5.0	194	0	194	237	44
3	5.0	176	0	359	204	-155
5	5.0	194	0	394	324	-70
9	6.5	357	312	44	15	-30
10	6.0	372	377	0	38	38
12	6.0	387	408	0	30	30
14	6.0	377	312	65	98	33
18	9.5	403	0	203	270	67
20	9.5	408	0	346	272	-74
21	9.5	366	0	166	300	134
22	6.0	286	0	450	425	-25

Table 3. Selected hours of the optimal day-ahead schedules (all 24 hours in results/table2_day_ahead_schedule.csv). Totals: deterministic 4,543 kWh, stochastic 4,782 kWh/day.

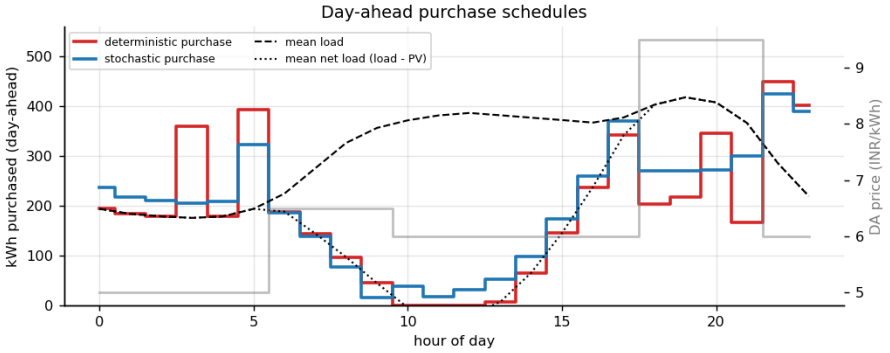


Figure 2. Day-ahead purchase schedules against mean load and mean net load.

The stochastic schedule **hedges where the mean forecast says no purchase is needed**: around midday (10–13 h) mean PV exceeds mean load, so the deterministic plan buys nothing, but on partly cloudy or overcast days PV falls far short and the shortfall would be bought in real time at twice the price; the stochastic plan therefore buys 17–45 kWh/h there. It also buys more in the 18–21 h peak (₹9.5/kWh, ₹19 in real time) and more in the cheap night hours to charge the battery. The total purchase is 5% higher (4,782 vs 4,543 kWh), the surplus on good days being resold at ₹3/kWh (536 vs 432 kWh/day): the price of insurance against expensive real-time purchases.

5.2 Expected cost, variability and risk versus the deterministic baseline

Model	Mean cost	Std dev	Median	95th pct	CVaR 90%	RT buy kWh/day	Resold kWh/day
Deterministic (mean-value)	34,043	7,017	31,504	48,528	49,184	405	432
Stochastic (SAA, S=300)	33,548	6,249	30,946	46,996	47,641	250	536

Table 4. Daily energy cost (₹) over 2,000 simulated out-of-sample days. CVaR 90% = mean cost of the worst 10% of days; RT = real-time purchases.

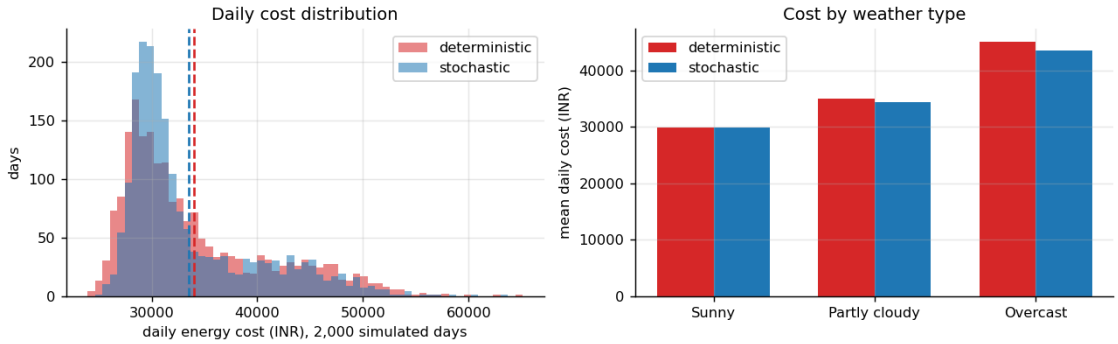


Figure 3. Left: daily-cost distributions. Right: mean cost by weather type.

The stochastic schedule has a lower mean (−₹495), a 11% lower standard deviation (₹6,249 vs ₹7,017), a lower 95th percentile (₹46,996 vs ₹48,528) and a lower worst-10%-day average (₹47,641 vs ₹49,184). It is cheaper on

63.3% of individual days. Unlike many hedging models, the benefit is therefore broad rather than a rare-event effect. Neither plan ever sheds load or curtails PV in the base case, because the grid backstop is always available: reliability is not at stake, price risk is.

Weather type	Test days	Cost, deterministic	Cost, stochastic	Saving ₹/day	Saving %
Sunny	1,044	29,969	29,858	111	0.37
Partly cloudy	628	35,025	34,443	582	1.66
Overcast	328	45,132	43,582	1,549	3.43

Table 5. Where the value comes from: cost by true weather type (out-of-sample).

Savings are small on sunny days (₹111, 0.4%) and large on overcast days (₹1,549, 3.4%): the mean-forecast plan is effectively a plan for a typical day that never occurs, and pays most when the actual day is far from typical.

5.3 Value of the stochastic solution and value of perfect information

Metric	Value
Stochastic expected daily cost (out-of-sample, N=2000)	33,548.26
Deterministic expected daily cost (EEV, out-of-sample)	34,043.15
VSS = EEV - RP (INR/day saved)	494.89
VSS 95% bootstrap CI lower	450.39
VSS 95% bootstrap CI upper	541.36
VSS as % of deterministic cost	1.45
Share of days stochastic plan is cheaper (%)	63.30
Wait-and-see expected cost (N=500)	30,137.94
EVPI = RP - WS (INR/day, N=500)	3,280.88
In-sample SAA objective (S=300)	33,499.42
Annual saving from VSS (INR/yr)	180,636.50

Table 6. Value metrics (₹/day unless a percentage). EVPI is computed on 500 of the test days.

VSS = ₹494.9/day (CI ₹450–541; 1.45% ≈ ₹180,637/year). **EVPI = ₹3,281/day** (6.6× the VSS): perfect weather and load forecasts would be worth much more than any improvement in the decision model, so investing in forecasting (satellite irradiance, numerical weather prediction) is the natural next step.

5.4 Convergence and stability of SAA

S	In-sample cost	Out-of-sample (training model)	Out-of-sample (true process)	Std of OOS cost	Std of total purchase	Optimism gap
10	33,438	34,065	33,628	197	341	627
25	33,619	33,867	33,475	74	209	248
50	33,778	33,864	33,472	89	230	86
100	33,706	33,820	33,405	56	174	115
200	33,785	33,814	33,408	56	167	29
400	33,795	33,784	33,379	36	80	-11

Table 7. SAA with S scenarios, 12 independent replications each (₹/day). For a cost-minimisation problem the in-sample value is biased low; optimism gap = out-of-sample (training model) − in-sample.

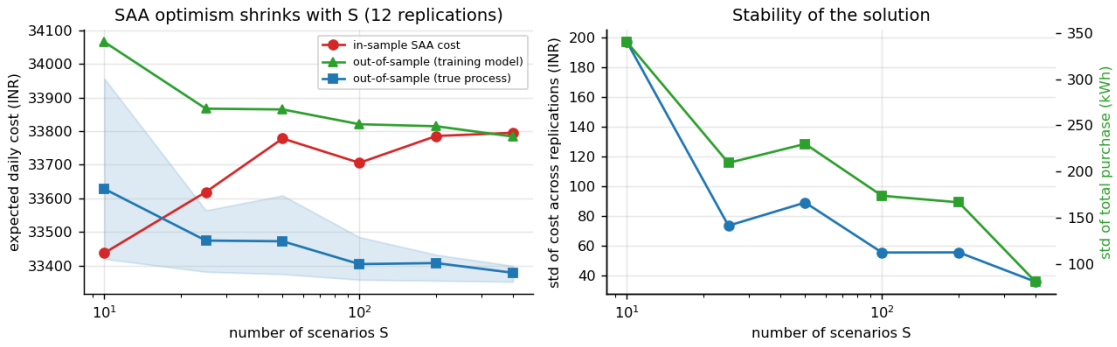


Figure 4. Left: in-sample optimism and out-of-sample cost versus S (band = 10–90% over replications). Right: variability of the solution across replications.

With S = 10 the in-sample cost is optimistic by ₹627 and the total purchase varies by ±341 kWh between replications. Out-of-sample cost on the true process falls from ₹33,628 (S = 10) to ₹33,379 (S = 400) and its replication-to-replication spread from ₹197 to ₹36, so the schedule becomes both better and more stable as S grows; the main results use S = 300, in the flat region.

5.5 Risk-averse (mean-CVaR) extension

λ	DA purchase kWh	Mean cost	Std dev	95th pct	CVaR 90%	Worst day
0.00	4,956	33,623	5,758	46,805	47,065	59,823
0.25	5,685	34,476	3,598	42,405	42,710	55,387
0.50	5,971	35,109	3,060	40,801	41,493	53,704
0.75	6,085	35,400	2,894	40,239	41,274	53,016
0.90	6,140	35,575	2,810	40,289	41,197	52,659

Table 8. Mean-CVaR frontier (S = 200, 1,000 out-of-sample days; ₹/day). λ = 0 is risk-neutral.

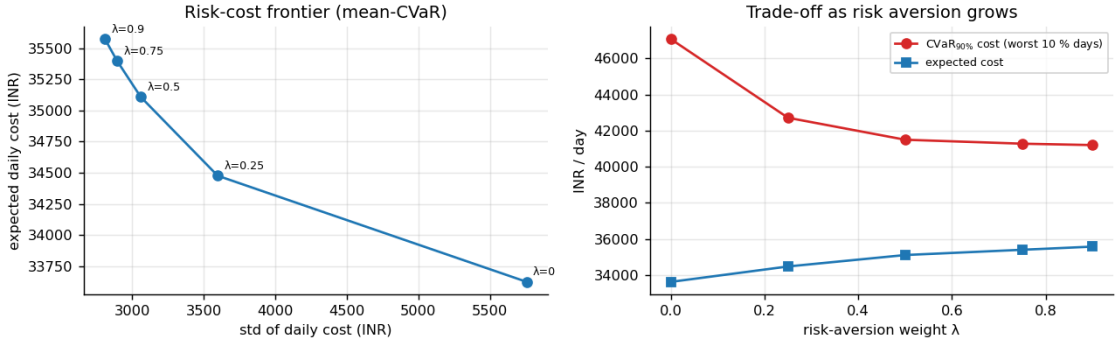


Figure 5. Risk–cost frontier: paying a little more on average removes much of the expensive tail.

λ = 0.25 costs ₹853/day more (2.5%) but cuts the standard deviation by 38% and CVaR from ₹47,065 to ₹42,710 (9%); λ = 0.5 reaches 47% lower volatility for +4.4% cost. Beyond λ ≈ 0.5 the frontier is almost flat, so a budget-constrained institution should choose λ between 0.25 and 0.5.

6. Sensitivity Analysis

6.1 Uncertainty drivers, price premium and storage

Factor	Value	Stoch. cost	Det. cost	VSS	VSS %
Overcast-day probability	0.05	32,236	32,572	337	1.03
Overcast-day probability	0.15	33,837	34,198	361	1.06
Overcast-day probability	0.25	34,600	35,157	557	1.59
Overcast-day probability	0.40	35,593	36,496	903	2.47
Weather variability multiplier	0.00	33,034	33,431	396	1.19
Weather variability multiplier	0.50	33,125	33,505	380	1.13
Weather variability multiplier	1.00	33,296	33,674	378	1.12
Weather variability multiplier	1.50	33,524	33,925	400	1.18
Weather variability multiplier	2.00	33,791	34,230	439	1.28
Real-time price multiplier	1.00	30,049	30,048	-1	-0.00
Real-time price multiplier	1.50	32,276	32,537	261	0.80
Real-time price multiplier	2.00	33,428	33,803	375	1.11
Real-time price multiplier	3.00	34,419	36,333	1,915	5.27
Real-time price multiplier	4.00	34,915	38,863	3,949	10.16
Battery size (kWh)	0.00	36,033	36,179	147	0.41
Battery size (kWh)	200.00	34,719	35,026	306	0.87
Battery size (kWh)	400.00	34,165	34,630	465	1.34
Battery size (kWh)	800.00	33,499	34,010	512	1.50
Battery size (kWh)	1,200.00	32,988	33,500	511	1.53

Table 9. One-factor sweeps in a controlled world (history of 365 days, S = 100 scenarios, 400 test days; cost in ₹/day).

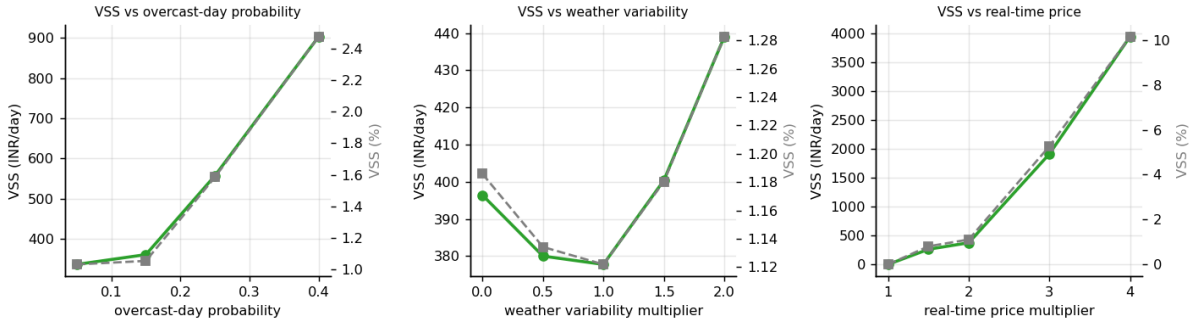


Figure 6. Value of the stochastic solution versus overcast frequency, within-state weather variability and real-time price multiplier.

The VSS is driven by the price asymmetry and by the weather mix, not by day-to-day noise. (i) At a real-time multiplier of 1× there is no penalty for deviating from the day-ahead schedule and the VSS is ≈ ₹0 (statistically zero), as theory predicts; it rises to ₹375 at 2× (base), ₹1,915 at 3× and ₹3,949 (10.2%) at 4×. (ii) More overcast days raise cost for both plans, and the VSS climbs from ₹337 (5% overcast) to ₹903 (40%). (iii) Scaling the within-state variability of κ from 0× to 2× barely moves the VSS (₹378–439), because the large uncertainty is *which weather state occurs*, which the scenarios retain. (iv) Storage is worth ₹2,534/day for an 800 kWh battery under the stochastic plan (cost ₹36,033 → ₹33,499); going to 1,200 kWh saves only a further ₹510/day, and the VSS saturates (₹512 → ₹511) once the battery is large enough to absorb most deviations.

6.2 Parameters (one-at-a-time ±20%)

Parameter (±20%)	Δ cost at −20%	Δ cost at +20%	Swing
Load level	-9,808	10,001	19,809
PV capacity	3,060	-2,933	5,994
Battery efficiency	1,228	-718	1,946
RT price multiplier	-821	585	1,406
Export price	263	-316	579
Battery size	244	-223	467
Battery wear cost	-142	142	284
Grid limit	130	-5	135
Value of lost load	0	0	0

Table 10. Change in expected daily cost (₹/day, plan re-optimised) when a parameter changes by ∓20%. Base expected cost in this world: ₹33,225.

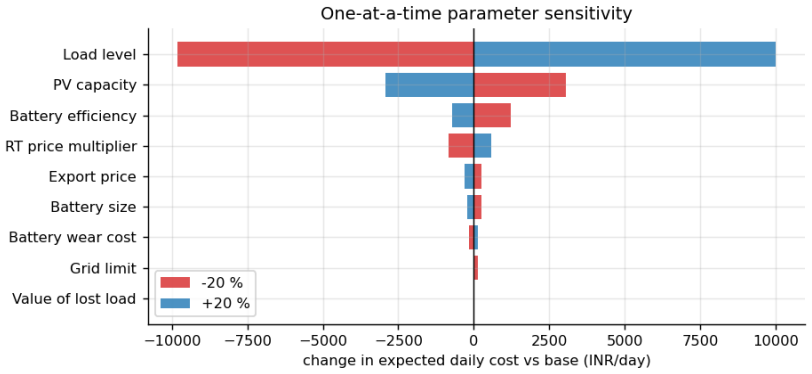


Figure 7. Tornado chart of parameter sensitivity.

Cost is dominated by the **load level** (±20% moves cost by about ±₹10,001/day) and by **PV capacity** (swing ₹5,994); battery round-trip efficiency (₹1,946) matters more than battery size (₹467). The value of lost load has **zero** effect because no scenario requires shedding when the grid backstop exists, an important structural insight: *the VOLL penalty could be omitted from this model without changing any decision.*

6.3 Robustness to distribution shift

True world (test)	Re-optimised (oracle)	Stochastic plan	Deterministic plan	Regret (stoch.)	Regret (det.)	Det. − Stoch.
Base (as trained)	32,829	32,844	33,304	15	475	460
Monsoon-heavy (35 % overcast)	34,857	34,911	35,581	55	724	670
Clear season (3 % overcast)	31,576	31,764	32,122	188	546	357
Load +10 %	37,995	38,810	40,347	815	2,352	1,537
Load -10 %	28,624	29,574	29,137	951	513	-438
Weather variability x1.5	33,622	33,607	34,138	-15	516	531

Table 11. Plans trained on the base history, tested in shifted worlds (₹/day, 600 days each). Regret = cost above the re-optimised oracle; negative 'Det. − Stoch.' means the deterministic plan wins.

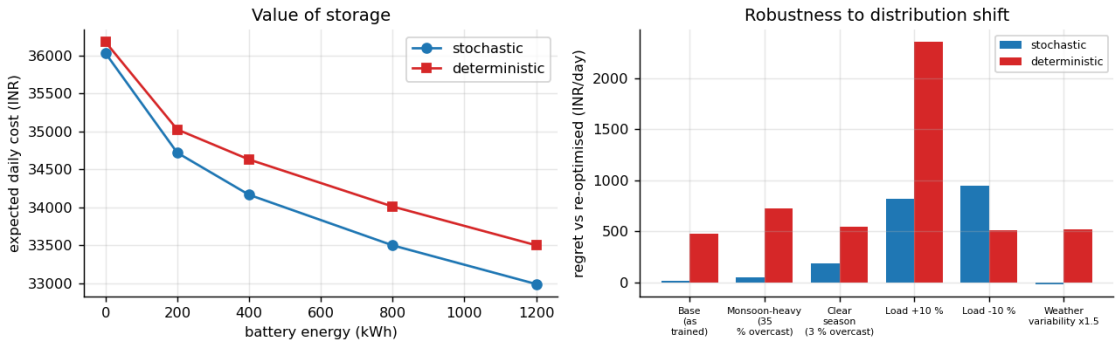


Figure 8. Left: value of storage. Right: regret under distribution shift.

The stochastic plan beats the deterministic one in five of six worlds, including a monsoon-heavy season (+₹670/day) and a +10% load increase (+₹1,537/day, where its regret of ₹815 is a third of the deterministic regret of ₹2,352). However, if load is **10% lower** than in the training data, the deterministic plan wins by ₹438/day: the stochastic plan has learned to over-buy as insurance and the surplus is resold at ₹3/kWh. Stochastic optimisation is only as good as its scenario set; a rolling re-estimation window (e.g. 60–90 days) is recommended.

6.4 Measured impact (SDG 9.5)

Indicator	Deterministic	Stochastic	Change
Net grid import (kWh/day)	4,516.44	4,497.23	-19.21
CO2 from grid import (kg/day, EF=0.71)	3,206.67	3,193.03	-13.64
PV energy curtailed (kWh/day)	0.00	0.00	0.00
Expected unserved energy (kWh/day)	0.00	0.00	0.00
Days with any load shedding (%)	0.00	0.00	0.00
Real-time (expensive) purchases (kWh/day)	404.82	250.45	-154.37
Surplus energy resold at low price (kWh/day)	431.66	535.61	103.95
Expected energy cost (INR/day)	34,043.15	33,548.26	-494.89

Table 12. Impact indicators per day, out-of-sample. CO_2 uses an assumed grid emission factor of 0.71 kg/kWh.

The stochastic plan reduces expensive real-time purchases by 38% (405 → 250 kWh/day) and expected cost by ₹495/day, while net grid import changes only marginally (-19 kWh/day, 5.0 t CO_2/year): the benefit is financial and price-risk reduction, not emissions, an honest and measurable result consistent with Target 9.5's focus on innovation capability.

7. Discussion, Limitations and Conclusions

Conclusions. (1) Modelling solar and load uncertainty explicitly cuts expected cost by ₹495/day (1.45%, significant by bootstrap) and volatility by 11%. (2) The gain arises from the asymmetry between real-time and resale prices and is concentrated on overcast days. (3) The value of the stochastic solution is driven by the real-time price premium and weather mix, not by within-state noise. (4) CVaR gives an explicit, cheap risk dial ($\lambda = 0.25$: −38% volatility for +2.5% cost). (5) SAA with ≥ 300 scenarios is accurate and stable. (6) EVPI ≈ 7× VSS, so better forecasting matters more than a better optimiser.

Limitations. Simulated data (absolute rupee amounts are illustrative); a single deterministic tariff and a flat real-time multiplier (real prices are also random); battery degradation is linear; no intra-hour dynamics; the plan is sensitive to a drop in load below the training level (Table 11). **Future work:** multi-stage rolling-horizon optimisation with forecast updates, stochastic prices, nonlinear battery ageing, distributionally robust formulations, and calibration with real smart-meter data.

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